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| **Fiscal Year:** | FY 2005 | **Task Last Updated:** | FY 01/10/2005 |
| **PI Name:** | Foley, Charles Michael Ph.D. | | |
| **Project Title:** | GABA Neurotransmission in Cardiovascular Function Following Simulated Microgravity in Male and Female Rats | | |
| **Division Name:** | Space Biology | | |
| **Program/Discipline:** | SPACE BIOLOGY | | |
| **Program/Discipline--Element/Subdiscipline:** | SPACE BIOLOGY--Organismal biology | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | (1) Cell & Molecular Biology<br>(2) Animal Biology: Vertebrate | | |
| **Space Biology Cross-Element Discipline:** | (1) Developmental Biology<br>(2) Musculoskeletal Biology<br>(3) Neurobiology | | |
| **Space Biology Special Category:** | None | | |
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| **City:** | Columbia | **State:** | MO |
| **Zip Code:** | 65211 | **Congressional District:** | 9 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 2003 Space Biology<br>03-OBPR-03 |
| **Start Date:** | 03/01/2004 | **End Date:** | 09/30/2006 |
| **No. of Post Docs:** | 0 | **No. of PhD Degrees:** | 0 |
| **No. of PhD Candidates:** | 0 | **No. of Master' Degrees:** | 0 |
| **No. of Master's Candidates:** | 0 | **No. of Bachelor's Degrees:** | 0 |
| **No. of Bachelor's Candidates:** | 0 | **Monitoring Center:** | NASA ARC |
| **Contact Monitor:** | | **Contact Phone:** | |
| **Contact Email:** | | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | NOTE: Changed end date to 9/30/2006 from original end date of 2/28/2007; NCE extension to 6/30/07 information was in error per A. Chu/ARC (8/07).<br>NCE to 6/30/07 per A. Chu 7/31/06 | | |
| **Key Personnel Changes/Previous PI:** | 0 | | |
| **COI Name (Institution):** | Hasser, Eileen M ( University of Missouri )<br>Heesch, Cheryl M ( University of Missouri ) | | |
| **Grant/Contract No.:** | NNA04CC62G | | |
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| **Performance Goal Text:** | | | |

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| **Task Description:** | Humans exposed to prolonged periods of microgravity undergo deconditioning of the cardiovascular system and exhibit increased incidence of orthostatic intolerance. In addition, the effects of cardiovascular deconditioning seem to be greater in females as compared to males that undergo periods of spaceflight. Hindlimb unloading (HU) in rodents is a well described animal model used to simulate exposure to microgravity in humans. HU rats exhibit signs of deconditioning including effects on the cardiovascular system that are very similar to those experienced by humans following spaceflight. The arterial baroreflex is a major component in the beat to beat maintenance of blood pressure and is critical for the normal compensation to an orthostatic stress. Following HU, baroreflex mediated increases in sympathetic nerve activity in response to hypotensive stimuli are attenuated, and this seems blunted to a greater extent in female rats following HU. This change in autonomic function is due in part to an increase in the effects of the inhibitory neurotransmitter gamma-aminobutyric acid (GABA) in a specific nucleus of the hindbrain, rostral ventrolateral medulla (RVLM), which is important in arterial baroreflex function and control of sympathetic activity. In addition, GABA(A) receptors are complex multimeric proteins that have a range of function and susceptibility to positive modulators is determined by their subunit composition. Taken together, the increased GABA neurotransmission in the RVLM following HU could be mediated by GABA receptors that are potentiated either due to changes in expression of individual subunits, presence of positive modulatory substances, enhanced susceptibility to endogenous modulators, increased number of GABA(A) receptors, greater GABA release, or a combination of these possibilities. The current proposal will test the general hypothesis that simulated microgravity alters GABA(A) neurotransmission in the RVLM, and to a greater extent in females, which accounts for the changes in autonomic function in both males and females.<br><br>The project has 4 specific objectives: 1. To examine the effects of HU on GABA(A) and GABA(B) receptor mediated inhibition in the RVLM in male and female rats. 2. To investigate alterations in GABA(A) receptor sensitivity to neurosteroids following HU in male and female rats. 3. To determine if HU in male and female rats alters nitric oxide neurotransmission and/or positive modulation of GABA(A) mediated inhibition in the RVLM. 4. To determine if known endogenous positive modulators of GABA(A) neurotransmission (nitric oxide and neurosteroids) contribute to autonomic changes that occur following HU in male and female rats.<br><br>Cardiovascular deconditioning results in reduced exercise capacity and orthostatic intolerance and these effects are more pronounced in women compared to men. Although a number of mechanisms likely are involved in the predisposition for orthostatic intolerance, dysfunction of cardiovascular sympathetic regulatory mechanisms appears to have an important contribution. Sex differences exist in arterial baroreflex control of the sympathetic nervous system and this reflex is altered following HU in male rats. Blunted baroreflex sympathoexcitation in male HU rats appears to be due to a central nervous system mechanism. Little is known concerning central mechanisms that may contribute to sex differences in cardiovascular deconditioning. Using the hindlimb unloaded rat model for cardiovascular deconditioning, the proposed studies will examine sex differences in GABA neurotransmission in the RVLM, the nucleus that is the major determinant of efferent sympathetic outflow. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | Control of the autonomic nervous system is altered during by exposure to microgravity or prolonged bedrest. This could contribute to the increased incidence of orthostatic hypotension and reduced ability to maintain blood pressure during hemorrhagic events following cardiovascular denconditioning. In addition, evidence suggests that females are more susceptible than males to these problems following spaceflight or prolonged bedrest. Defining the central nervous system mechanisms that account for the changes in autonomic nervous system function following cardiovascular deconditioning will increase our understanding of the adaptations that occur in males and females following spaceflight. These studies utilize a ground base model of cardiovascular deconditioning. Therefore, the knowledge gained from these studies would apply not only to exposure to midcrogravity but also to patients exposed to prolonged bedrest and to normal neural control of the circulation. |
| | Progress has been made on several difference research projects investigating the cardiovascular changes following HU. These include studies in our laboratory and collaborative projects with other investigators. We have developed an alternative method to the Morey-Holton method to hindlimb suspend rats. In addition, we have initiated studies investigating the effects of sex and HU both on the mRNA expression of GABA(A) receptor subunit expression in the RVLM and on GABA(A) and GABA(B) neurotransmission in the RVLM. Furthermore, we have contributed to several joint projects with other researches.<br>Successful adaptations to the traditional hindlimb suspension model in the rat.<br><br>The purpose of this study was to evaluate an alternative method to the Morey-Holton technique for hind limb unloading (HU) that would permit tail examination during long-term suspension, surgery during unloading and a seamless recovery phase following HU. The proposed method also provides a solution to the problem of HU rats chewing the traction tape and coming out of the suspension apparatus. Data are presented from two separate laboratories currently using the new technique. Young (9 weeks) male and female (n=12) and mature (6 and 16 months) female rats (n=12) were HU for 2 or 4 weeks with or without 2 weeks of recovery. Two stainless steel wires were passed through separate intervertebral disc spaces at the level of the ~5th and 7th sacral vertebrae and fashioned into a ring from which animals were suspended. This position was chosen as it is not too proximal to interfere with defecation and not too distal where vertebrae are too small to hold body weight. All HU groups had ambulatory controls, some of which were pair-fed. Adrenal weights, daily food consumption, body weight lost during suspension and atrophy of the soleus muscle were used to determine success of hind limb suspension. Body mass was not significantly different at baseline and no weight loss occurred after unloading in any of the groups. Soleus muscle mass was significantly reduced with unloading ($p<0.001$) in all groups. Food consumption tended to be higher in the HU groups compared to controls and thus, there was no food restriction of controls. Adrenal gland weights did not differ between controls and any of the HU rats. The lack of significant change in body mass, food consumption, adrenal gland weight, coupled with significant soleus muscle atrophy, suggest this method is as successful as the traditional Morey-Holton technique. The method described confers several advantages such as ease of reloading and elimination of rats chewing the traction tape and becoming ambulatory. The disadvantage to the technique is that rats must be anesthetized for 5-7 minutes to implant tail rings.<br><br>mRNA expression of GABAA receptor subunits in the RVLM.<br><br>The purpose of these studies is to examine the effects of hindlimb unloading (HU) on mRNA expression levels |

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| **Task Progress:** | GABA(A) receptor subunits in the RVLM in male and female rats. Previous studies have shown that GABA(A) mediated inhibition in the RVLM is increased following HU in male rats and this could be due to increased positive modulation by neurosteroids. Neurosterods are metabolites of progesterone and are potent endogenous positive modulators of GABA(A) receptor function. The neurosteroids directly bind to the receptor and facilitate the actions of GABA. The ability of neurosteroids to positively modulate GABA(A) receptors is dependent on the subunit composition of the GABA(A) receptors. Following HU, enhancement of GABA(A) receptor function due to neurosteroids may be greater in females than males. Potentiation of GABA(A) receptors by neurosteroids is consistent with enhanced GABA(A) mediated inhibition in the RVLM following HU and may be a mechanism that accounts for the greater reduction in baroreflex mediated sympathoexcitation. These studies will evaluate changes in expression in GABA(A) receptor subunits important in determining neurosteroid sensitivity. To date, we have developed the standards and optimized the real time reverse transcriptase-polymerase chain reaction (RT-PCR) necessary to quantify mRNA expression of glyceraldehydes 3-phosphate dehydrogenase (GAPDH) and GABA(A) alpha1, alpha2, gamma1, gamma2, gamma3, and delta receptor subunits. We are currently working on cloning and producing the standards for GABAA alpha3, alpha4, alpha5, and alpha6 receptor subunits. With the development of these techniques, we can now investigate differences in the expression of specific GABA(A) receptor subunits in the RVLM between male and female rats and those produced by a period of cardiovascular deconditioning.<br><br>Role of sex and HU on GABA(A) and GABA(B) neurotransmission in the RVLM.<br><br>GABA has a large, tonic inhibitory influence on the RVLM, and this is mediated via GABA(A) and GABA(B) receptors. Previous studies have shown that GABAA mediated inhibition in the RVLM is increased following HU in male rats. This could be due to changes in receptor subunit composition and sensitivity, increases in positive modulation by NO or neurosteroids, or alterations in GABAergic neuronal inputs to the RVLM. In addition to GABA(A) mediated inhibition, HU may increase GABA(B) neurotransmission in the RVLM which could contribute to the altered baroreflex control of sympathetic nerve activity observed following HU. Importantly, a greater potentiation of GABA(A), GABA(B) or both mediated inhibition of the RVLM in females than males could account for the greater reduction in baroreflex mediated sympathoexcitation. These experiments begin to investigate GABA(A) and GABA(B) mediated inhibition in the RVLM following HU in male and female rats. Specifically, these experiments evaluate the sensitivity of GABA(A) and GABA(B) receptors and the interaction of these two receptor systems. Bilateral microinjection of the GABA(A) antagonist bicuculline or the GABA(B) antagonist CGP-35348 into the RVLM increased MAP, heart rate, and lumbar sympathetic nerve activity (LSNA) in male and female rats that were either control and had undergo a 14 day HU period. These data are consistent with previous studies and suggest that both GABA(A) and GABA(B) receptors are tonically active, inhibit the RVLM, and are involved in resting sympathetic nerve activity. Unilateral microinjection of a GABA(B) agonist baclofen decreased MAP, heart rate, and LSNA. To date, the number of animals in each group is low (n=2-4 per group) and does not allow for meaningful comparison. These experiments are in progress and their completion will assess if there are differences in tonic GABA(A) and GABA(B) mediated inhibition at the RVLM between male and female rats and if it is effected by cardiovascular deconditioning. |
| **Bibliography Type:** | Description: (Last Updated: 08/28/2007) |
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